



Time to read: 5–7 minutes

Title: *AI Landscape Today: Industries and Problem Types*

Big idea: AI projects are driven by business problems, not algorithms.

Why this section matters?

Many teams start AI conversations in the wrong place: “*Should we use deep learning?*” “*Should we use an LLM?*” That creates confusion, wasted effort and projects that never reach production.

A better sequence is:

Problem → Decision → Data → System → Model

- **Problem:** What pain exists (cost, risk, downtime, customer dissatisfaction)?
- **Decision:** What decision will improve if we automate or assist it?
- **Data:** Do we have signals that reflect that decision?
- **System:** How will this integrate into workflows?
- **Model:** Only then do we choose a technique.

This is why understanding the landscape matters: it trains you to think in problems and patterns, not buzzwords.

1) AI Across Industries (What AI does in the real world)

AI is not “one thing.” In practice, AI is used where decisions are:

- High-volume (too many for humans)
- Repetitive (same type of decision over and over)
- Time-sensitive (need quick action)
- Costly if wrong (fraud, safety, downtime)
- or complex (too many factors for manual reasoning)

Below are common industries and the kinds of decisions AI supports.

Healthcare

Where AI is used?

- Diagnosis support (help doctors interpret evidence faster)
- Medical imaging classification (detect patterns in X-rays/MRI/CT)
- Patient risk prediction (who is at high risk of complications)
- Operational efficiency (bed utilisation, scheduling)

Examples

- Predicting disease risk
- Classifying medical images
- Optimising hospital resource usage

Explanation (Why AI fits here)

Healthcare has high stakes and high complexity. Humans are skilled, but the workload is large and decisions depend on many signals. AI is often used as **decision support**, not full automation.

Key point: The goal is better outcomes and safer workflows—not “fancy models.”

Finance

Where AI is used?

- Fraud detection (spot unusual patterns quickly)
- Credit scoring / default risk (estimate risk with available signals)
- Transaction monitoring (detect suspicious behaviour for compliance)

Examples

- Fraud detection
- Credit scoring
- Transaction monitoring

Explanation

Finance is about **risk and trust**. AI helps by detecting patterns humans can't easily see at scale. But errors have costs:

- **False negatives** (missed fraud) → direct loss
- **False positives** (blocking legitimate users) → churn, reputational damage.

Key point: You don't optimise accuracy alone—you optimise business risk.

Manufacturing

Where AI is used?

- Predictive maintenance (forecast failure before it happens)
- Quality inspection (detect defects in images/sensor signals)
- Supply chain optimisation (reduce delays, inventory waste)

Examples

- Predictive maintenance
- Quality inspection
- Supply chain optimisation

Explanation

Manufacturing produces continuous streams of sensor and operational data. Failures and defects are expensive, and many patterns show up **before** the event. AI is valuable because it can convert noisy signals into early warnings.

Key point: Small improvements can save millions because downtime is extremely costly.

Retail

Where AI is used?

- Product recommendations (increase conversion and engagement)
- Demand forecasting (stock the right quantity at the right time)
- Pricing optimisation (balance margin, demand, competition)

Examples

- Product recommendations

- Demand forecasting
- Pricing optimisation

Explanation

Retail decisions must be made at huge scale—thousands of products, millions of customers, frequent changes. Often, a good-enough model that is fast and scalable beats a perfect model that is slow and expensive.

Key point: In retail, **speed and scale matter as much as accuracy.**

IT Services

Where AI is used?

- Automation (ticket routing, summarisation, knowledge retrieval)
- Incident prediction (early warning from logs and metrics)
- Intelligent monitoring (detect anomalies and reduce noise)
- Developer tools (code generation, testing support, productivity)

Examples

- Automation
- Incident prediction
- Intelligent monitoring
- Developer tools

Explanation

This is where AI meets software engineering directly. IT environments produce massive telemetry data (logs, traces, metrics). AI helps reduce manual effort and improves response times—but it must be reliable and observable.

Key point: This domain naturally leads into MLOps because production integration is the core challenge.

2) AI Problem Types (Patterns that repeat across industries)

Different industries use different data, but the **problem shapes** repeat. Recognising these problem types helps you choose the right approach and evaluation metrics.

Prediction

Question: *What is likely to happen next?*

Examples

- Demand forecasting
- Failure prediction
- Risk scoring

Explanation

A prediction typically outputs a **number** (score, probability, forecast). It supports planning and proactive action.

Classification

Question: *Which category does this belong to?*

Examples

- Spam vs not spam
- Fraud vs non-fraud
- Defective vs non-defective

Explanation

Classification outputs a **label** (class). It is used for routing decisions such as approve/reject, pass/fail and safe/unsafe.

Recommendation

Question: *What should we show or suggest next?*

Examples

- Products
- Content
- Actions

Explanation

Recommendation outputs a **ranked list**. Success is often measured not by accuracy, but by business outcomes such as conversion, engagement and retention.

Optimisation

Question: *What is the best decision under constraints?*

Examples

- Route optimisation
- Inventory management
- Resource allocation

Explanation

Optimisation problems include constraints (time, cost, capacity). AI may predict outcomes, but the system also needs decision logic to pick the best action.

Generation

Question: *What new content can we produce?*

Examples

- Text generation
- Image generation
- Code generation

Explanation

Generation is where foundation models (Gen AI) are dominant. The output is content rather than a label. Evaluation is harder—often using human feedback, quality rubrics, and safety constraints.

3) The Most Important Message (Read carefully)

AI projects are driven by business problems, not algorithms.

We don't start by saying:

- “Let's use a neural network.”
- “Let's use an LLM.”

We start by asking:

- **What problem are we solving?**
- **Who is affected?**
- **What decision will this improve?**

Because:

The model is a tool. The system is the solution.

Explanation (why this matters)

A great model that sits in a notebook creates no value.

A slightly imperfect model integrated into a workflow, monitored and continuously improved can create significant value.

This course focuses on making you capable of building that second kind of system.

Closing bridge to the next topic

Once you understand the problem and the type of AI work, you can choose the right lifecycle, tools and deployment strategy.

That's why next we'll learn **how AI systems are built differently from traditional software**—and why AI needs an AI-specific SDLC.